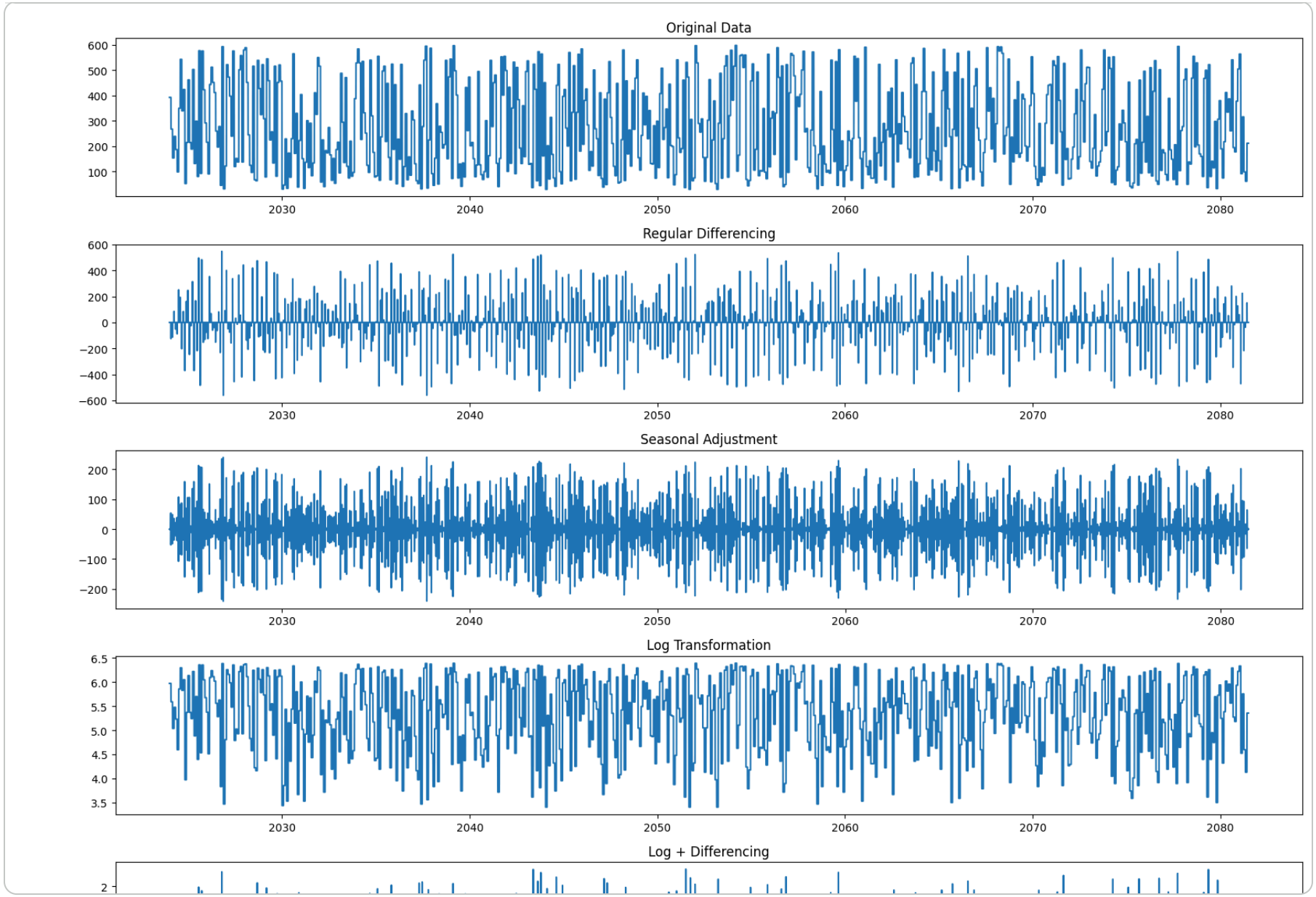
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose


data = pd.read_csv("/content/user_behavior_timeseries.csv")

data.columns = data.columns.str.strip()


data['Date'] = pd.date_range(start='2024-01-01', periods=len(data))
data['Date'] = pd.to_datetime(data['Date'])


data.set_index('Date', inplace=True)


col = "App Usage Time"

data['usage_diff'] = data[col] - data[col].shift(1)


result = seasonal_decompose(data[col], model='additive', period=7)
data['usage_sea_diff'] = result.resid

data['usage_log'] = np.log(data[col])


data['usage_log_diff'] = data['usage_log'] - data['usage_log'].shift(1)


result = seasonal_decompose(data['usage_log_diff'].dropna(), model='additi
data['usage_log_seasonal_diff'] = result.resid


plt.figure(figsize=(16, 16))
```

```
plt.subplot(6,1,1)
plt.plot(data[col])
plt.title("Original Data")

plt.subplot(6,1,2)
plt.plot(data['usage_diff'])
plt.title("Regular Differencing")


plt.subplot(6,1,3)
plt.plot(data['usage_sea_diff'])
plt.title("Seasonal Adjustment")

plt.subplot(6,1,4)
plt.plot(data['usage_log'])
plt.title("Log Transformation")

plt.subplot(6,1,5)
plt.plot(data['usage_log_diff'])
plt.title("Log + Differencing")

plt.subplot(6,1,6)
plt.plot(data['usage_log_seasonal_diff'])
plt.title("Log + Seasonal Differencing")

plt.tight_layout()
plt.show()
```


Original Data
600
500
400
300
200
100
2030
2040
2050
2060
2070
2080
Regular Differencing
600
400
200
0
−200
−400
−600
2030
2040
2050
2060
2070
2080
Seasonal Adjustment
200
100
0
−100
−200
2030
2040
2050
2060
2070
2080
Log Transformation
6.5
6.0
5.5
5.0
4.5
4.0
3.5
2030
2040
2050
2060
2070
2080
Log + Differencing
2